

Teaching Exam

Subject – Computer Science

Topic – Practice and Revision

Lecture No.- 04



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Topics to be Covered

Topic

1:- Practice & Revision Set



1. If A and B are the inputs of a half adder, the sum is given by _____

- a) A AND B
- b) A OR B
- c) ☒ A XOR B
- d) A EX-NOR B

2. Half-adders have a major limitation in that they cannot _____

- a) Accept a carry bit from a present stage
- b) Accept a carry bit from a next stage
- c) ☒ Accept a carry bit from a previous stage
- d) Accept a carry bit from the following stages

Handwritten diagrams illustrating the limitation of a half-adder:

$$\begin{array}{r} 1 \\ 1 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 0 \\ 1 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 1 \\ 0 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 0 \\ 0 \\ \hline 0 \end{array}$$

1. Answer: c

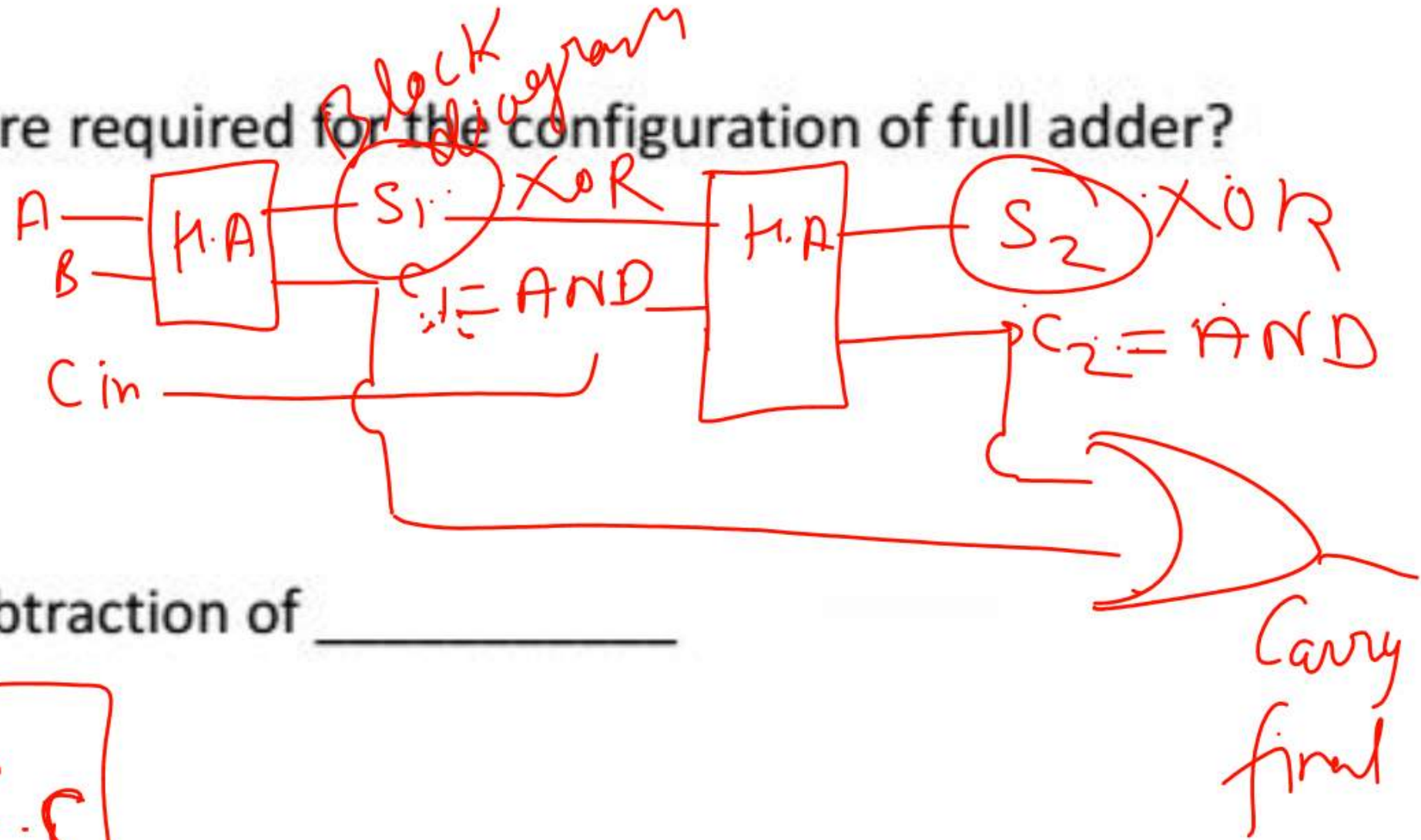
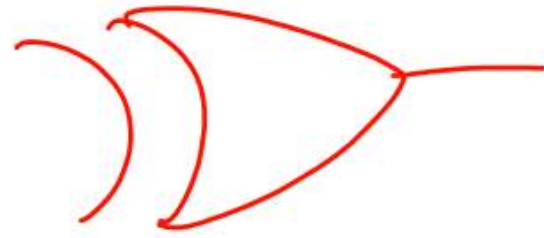
Explanation: If A and B are the inputs of a half adder, the sum is given by $A \oplus B$, while the carry is given by $A \text{ AND } B$.

2. Answer: c

Explanation: Half-adders have a major limitation in that they cannot accept a carry bit from a previous stage, meaning that they cannot be chained together to add multi-bit numbers. However, the two output bits of a half-adder can also represent the result $A+B=3$ as sum and carry both being high.

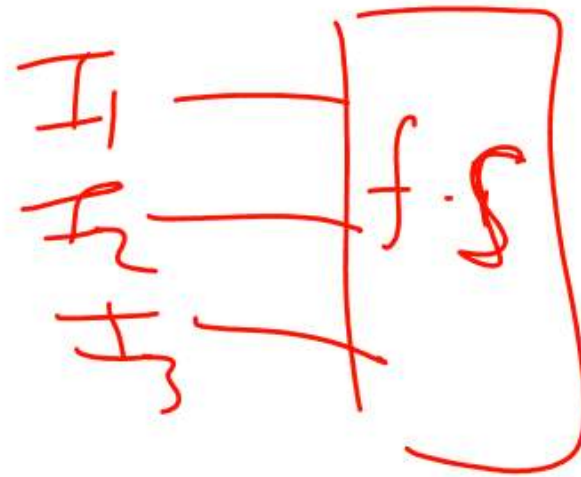
3. How many AND, OR and EXOR gates are required for the configuration of full adder?

- a) 1, 2, 2
- b) 2, 1, 2
- c) 3, 1, 2
- d) 4, 0, 1



4. Full subtractor is used to perform subtraction of _____

- a) 2 bits
- b) 3 bits
- c) 4 bits
- d) 8 bits



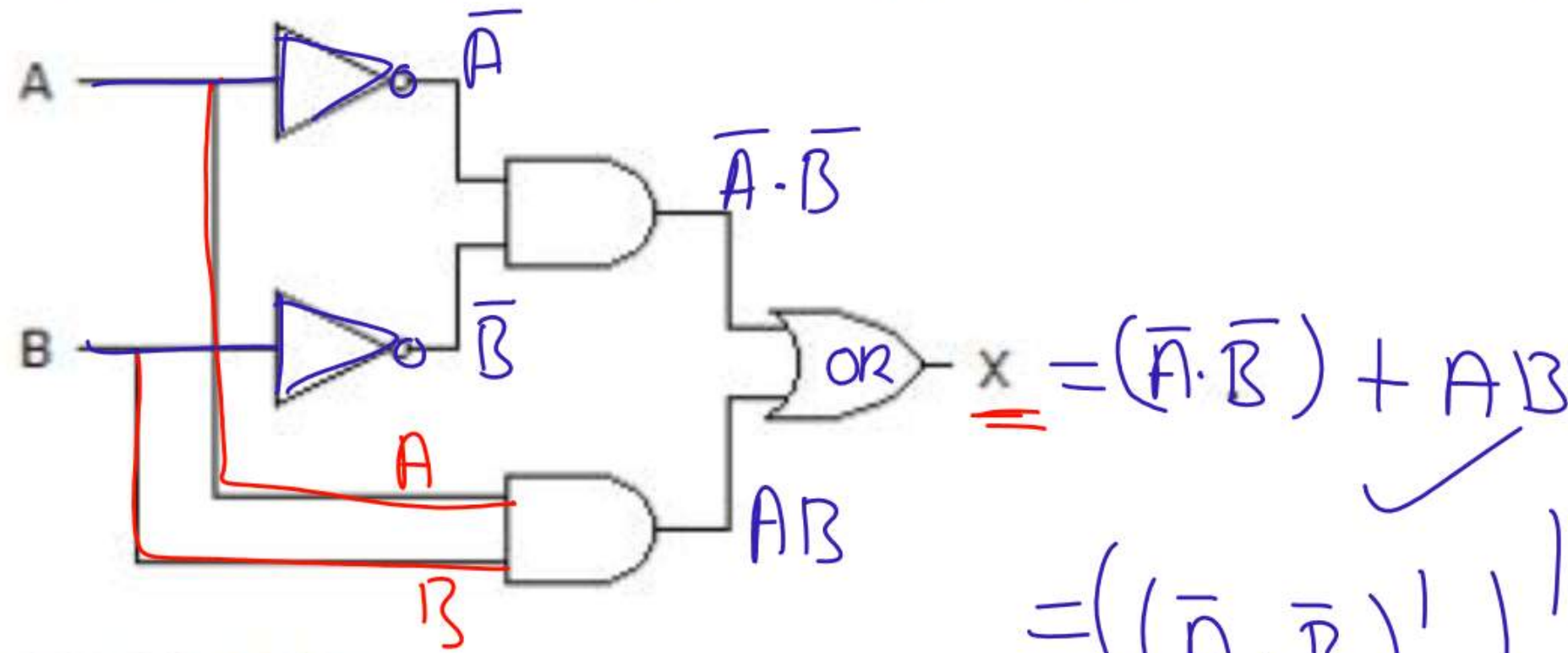
3. Answer: b

Explanation: There are 2 AND, 1 OR and 2 EXOR gates required for the configuration of full adder, provided using half adder. Otherwise, configuration of full adder would require 3 AND, 2 OR and 2 EXOR.

4. Answer: b

Explanation: Full subtractor is used to perform subtraction of 3 bits, namely minuend bit, subtrahend bit and borrow from the previous stage. However, it also produces 2 outputs BORROW and DIFFERENCE..

2. Which of the following logic expressions represents the logic diagram shown?



- a) $X = AB' + A'B$
- b) $X = (AB)' + AB$
- c) $X = (AB)' + A'B'$
- d) $X = A'B' + AB$

5. Answer: d

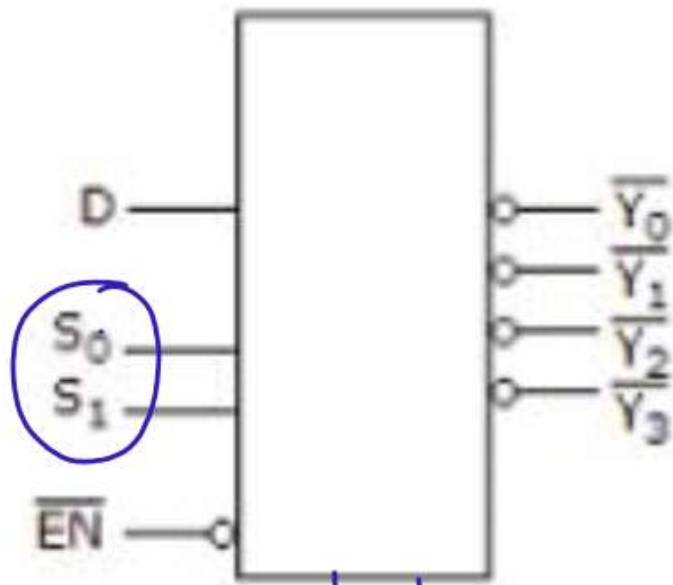
Explanation: 1st output of AND gate is $= A'B'$

2nd AND gate's output is $= AB$ and,

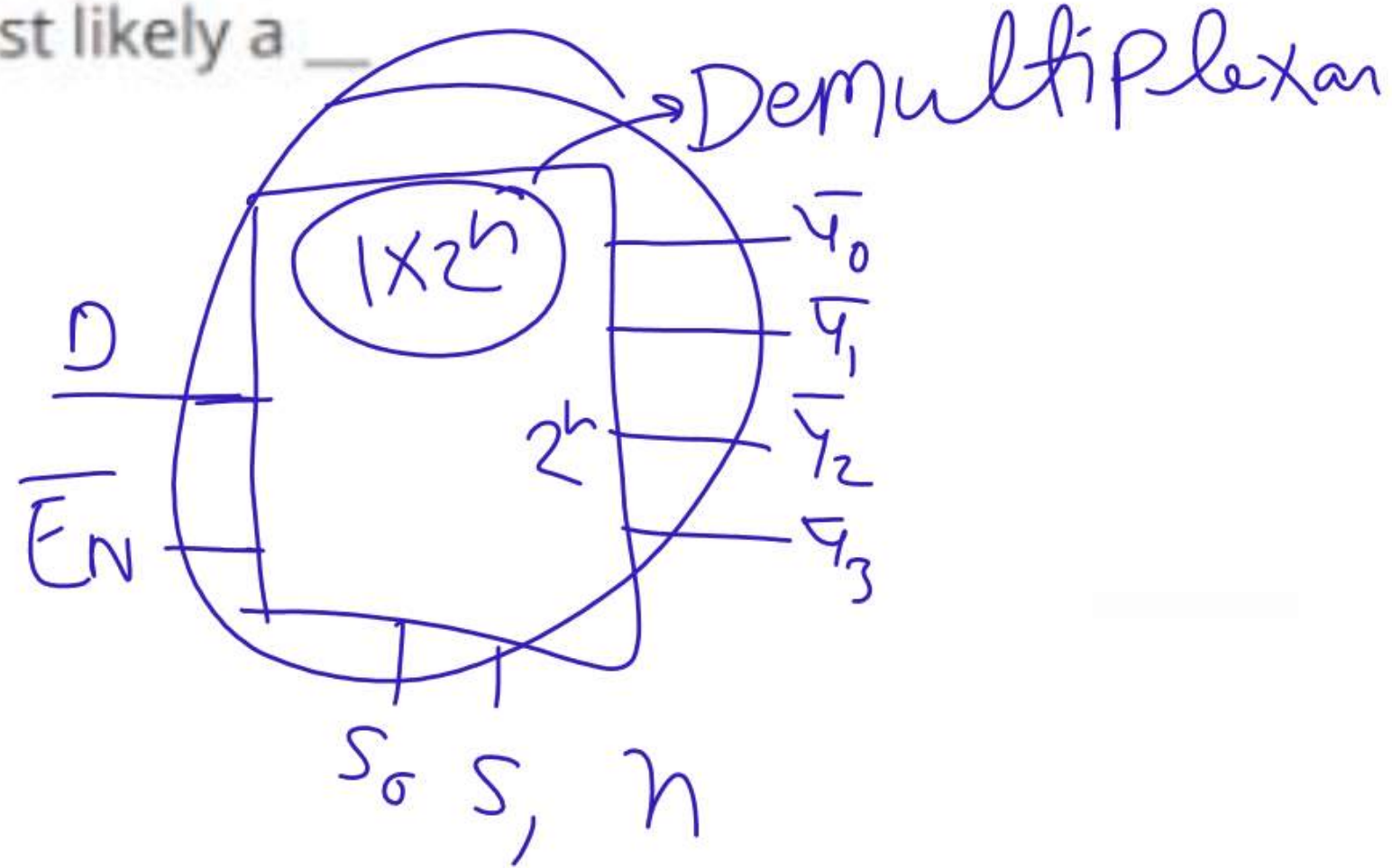
OR gate's output is $= (A'B') + (AB) = AB + A'B'$.

6.

6. The device shown here is most likely a —

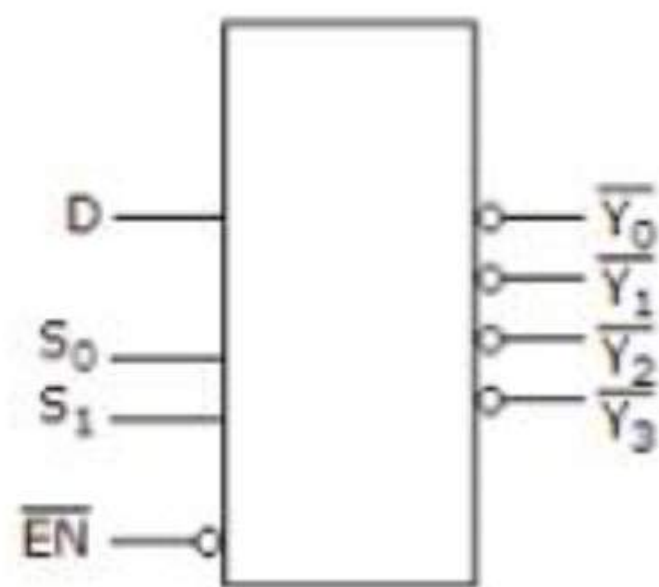


- a) Comparator
- b) Multiplexer
- c) Inverter
- d) Demultiplexer



6.

3. The device shown here is most likely a __

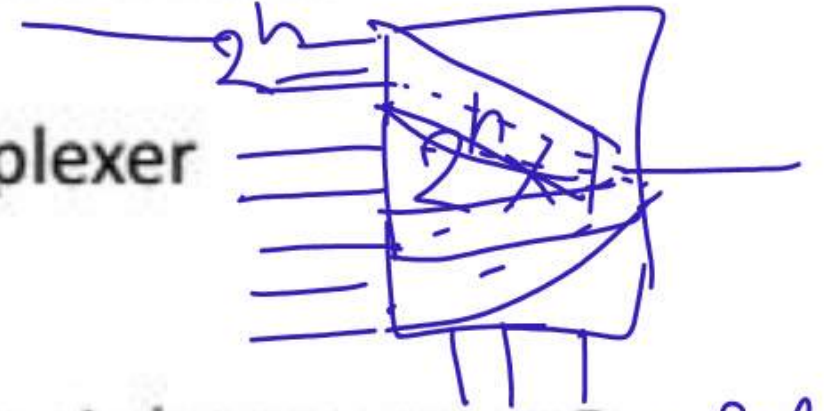


- a) Comparator
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6. Answer: d

Explanation: The given diagram is demultiplexer, because it takes single input & gives many outputs. A demultiplexer is a combinational circuit that takes a single output and latches it to multiple outputs depending on the select lines.

7. What is data routing in a multiplexer?
- a) It spreads the information to the control unit
 - b) It can be used to route data from one of several source to destination
 - c) It is an application of multiplexer
 - d) It can be used to route data and it is an application of multiplexer



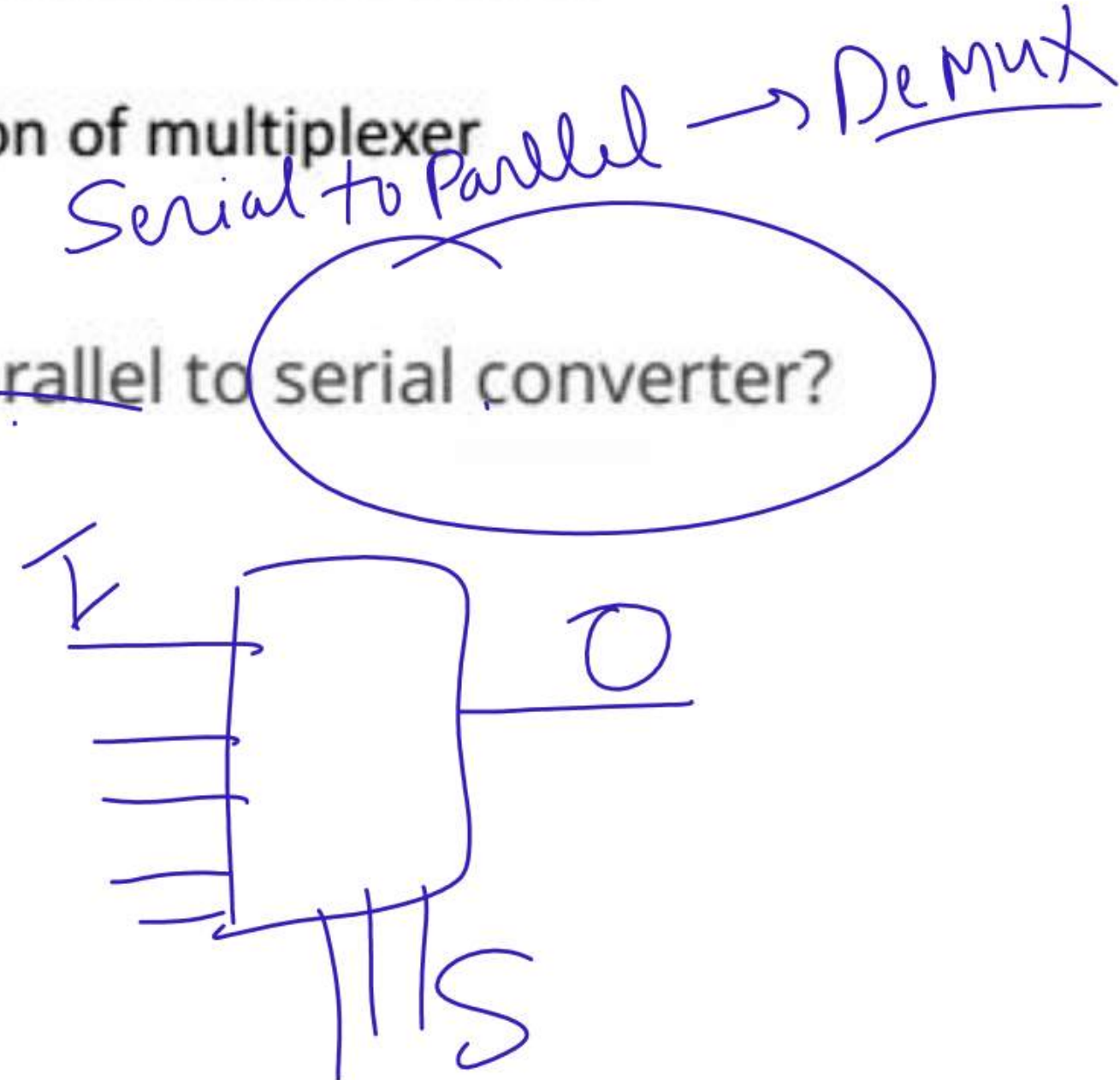
8 .Which of the following circuit can be used as parallel to serial converter? , 1 Sel,

- a) Multiplexer
- b) Demultiplexer
- c) Decoder
- d) Digital counter

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8. Which of the following circuit can be used as parallel to serial converter?

- a) Multiplexer
- b) Demultiplexer
- c) Decoder
- d) Digital counter



7. Answer: d

Explanation: Multiplexing means passing more than one data through the same channel. Data routing is an application of multiplexer and it can be used to route data from one of several source to destination.

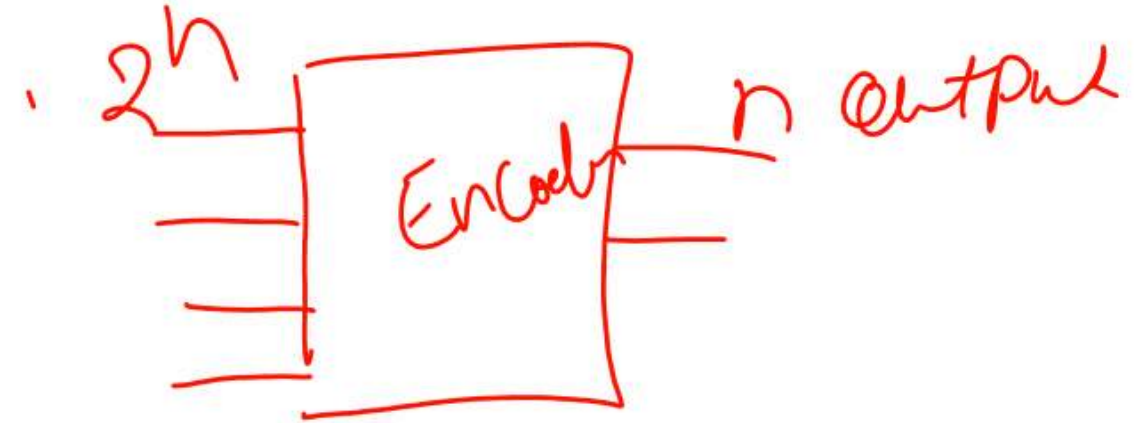
8 . Answer: a

Explanation: A combinational circuit that selects one from many inputs is known as Multiplexer. In multiplexer, different inputs are inserted parallelly and then it gives one output which is in serial form.

9. Can an encoder be a transducer?
- a) ~~Yes~~
 - b) No
 - c) May or may not be
 - d) Both are not even related slightly

- 10 If two inputs are active on a priority encoder, which will be coded on the output?
- a) The higher value
 - b) The lower value
 - c) Neither of the inputs
 - d) Both of the inputs

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 - b) The lower value
 - c) Neither of the inputs
 - d) Both of the inputs

9. Answer: a

Explanation: Of course, a transducer is a device that has the capability to emit data as well as to accept. Transducer converts signal from one form of energy to another.

10 Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. If two inputs are active on a priority encoder, the input of higher value will be coded in the output

11. who created the first DBMS?

a) Edgar Frank Codd

b) Charles Bachman

c) Charles Babbage

d) Sharon B. Codd.

ER Model

12. Which type of data can be stored in the database?

a) Image oriented data

b) Text, files containing data

c) Data in the form of audio or video

d) All of the above

11. Answer: b

Explanation: Charles Bachman along with his team invented the first DBMS known as Integrated Data Store (IDS).

12. Answer: d

Explanation: The reason for creating the database management system was to store large data and these data can be of any form image, text, audio, or video files, etc. DBMS allows the users to store and access the data of any format.

11. Answer: b

Explanation: Charles Bachman along with his team invented the first DBMS known as Integrated Data Store (IDS).

12. Answer: d

Explanation: The reason for creating the database management system was to store large data and these data can be of any form image, text, audio, or video files, etc. DBMS allows the users to store and access the data of any format.

13. Which of the following is a component of the DBMS?

- a) Data ✓
- b) Data Languages ✓
- c) Data Manager
- d) All of the above ✓

14. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

- a) Relation set
- b) Tuples
- c) Entity set
- d) Entity Relation model.

13. Answer: d

Explanation: The components of DBMS are as follows:

1) ~~Hardware~~: Like a hard drive, monitor, etc.

2) ~~Software~~: Provides a user interface

3) ~~Data Manager~~: Manages operations of DBMS.

4) ~~Data~~: The collection of information on the DB is known as data.

5) Data Languages: Languages like DDL, DML, ^{TCL}~~DAL~~, and DCL allow to perform operations like creating, modifying, storing, or retrieving data.

14. Answer: c

Explanation: In the actual world, an entity is a distinct “thing” or “object” from all other objects. For example: Each employee of an organization is an entity.

15. What does an RDBMS consist of?

- a) Collection of Records
- b) Collection of Keys
- c) ~~Collection of Tables~~
- d) Collection of Fields.

16. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____ integrity constraint.

- a) ~~Referential~~
- b) Primary
- c) Referencing
- d) Specific

15. Answer: c

Explanation: It consists of a collection of tables i.e., the data is organized in tabular format. The columns of the relation are known as Fields and rows of the relation are known as Records. Constraints in a relation are known as Keys.

16. Answer: a

Explanation: Consider 2 relations r_1 and r_2 . r_1 may include among its attributes the primary key of relation r_2 . This attribute is called a foreign key from r_1 , referencing r_2 . The relation r_1 is also called the referencing relation of the foreign key dependency, and r_2 is called the referenced relation of the foreign key.

17. The ability to query data, as well as insert, delete, and alter tuples, is offered by _____
- a) TCL (Transaction Control Language)
 - b) DCL (Data Control Language)
 - c) DDL (Data Definition Language)
 - d) DML (Data Manipulation Language).
18. _____ is a set of one or more attributes taken collectively to uniquely identify a record.
- a) Primary Key
 - b) Foreign key
 - c) Super key
 - d) Candidate key

17. Answer: d

Explanation: A query is a request for data or information. Relational Schema is the design and structure of the relation. DDL consists of commands that help in modifying. DML performs the change in the values of the relation..

18. Answer: c

Explanation: Foreign key creates a relationship between two relations. Super key is the superset of all the keys in a relation. A candidate key is used to identify tuples in a relation.

19. Which command is used to remove a relation from an SQL?
- a) ~~Drop table~~
 - b) Delete
 - c) Purge
 - d) Remove
20. Which of the following command is correct to delete the values in the relation teaches?
- a) ~~Delete from teaches;~~
 - b) Delete from teaches where Id = 'Null';
 - c) Remove table teaches;
 - d) Drop table teaches;

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21. Procedural language among the following is _____

- a) Domain relational calculus
- b) Tuple relational calculus
- c) Relational algebra
- d) Query language

22. _____ operations do not preserve non-matched tuples.

- a) Left outer join
- ☒ b) Inner join
- c) Natural join
- d) Right outer join

21. Answer: c

Explanation: Non-Procedural Languages are Domain relational calculus and Tuple relational calculus. Relational algebra is a procedural language that takes input in the form of relation and output generated is also a relation

22. Answer: b

Explanation: Left outer join returns all the rows from the table that is on the left side and matching rows on the right side of the join. Inner join returns all rows when there is at least one match in BOTH tables. Natural join returns the common columns from the tables being joined. A right outer join returns all the rows from the table that is on the right side and matching rows on the left side of the join.

23. Which forms have a relation that contains information about a single entity?

a) 4NF

b) 2NF

c) 1NF

d) 3NF

3NF = 2NF + ^{NO} Transitive Dependency

24. The top level of the hierarchy consists of _____ each of which can contain Schema.

a) Schemas, Catalogs

b) Schemas, Environment

c) Environment, Schemas

d) Catalogs, Schemas

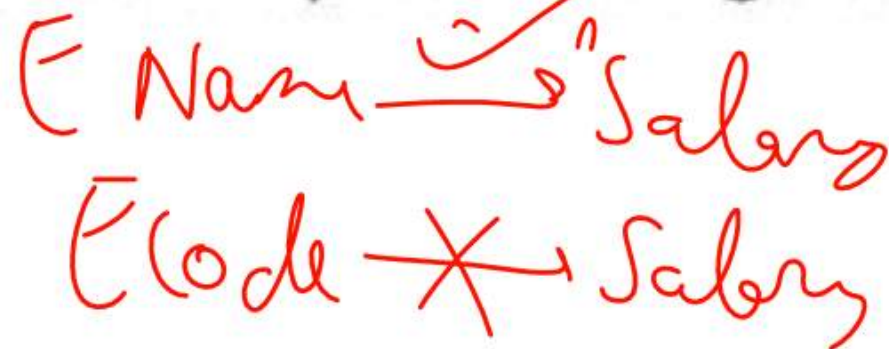
23. Answer: d

Explanation: A table is in its 3rd NF if it has already completed in 2NF, and non-key attributes are not dependent on any non-key attributes. This means that all non-key attributes must depend on primary key. When a table represents a single entity, the primary key will identify the instance row and all attributes in 3NF are dependent on the primary key. So, the primary key in the 3th NF captures the whole information of this single entity



24. Answer: d

Explanation: Schemas represent the logical configuration of the DBMS. Catalogs consist of metadata of the objects and system settings used.



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25

Why the following statement is erroneous?

```
SELECT dept_name, ID, avg (salary)
FROM instructor
GROUP BY dept_name;
```

Handwritten annotations: A red circle around 'dept_name' with 'OK' above it. A red circle around 'ID' with an 'X' above it. A red circle around 'avg (salary)'.

- a) ☒ Dept_id should not be used in group by clause
- b) ☐ Group by clause is not valid in this query
- c) ☐ Avg(salary) should not be selected
- d) ☐ None

26. After groups have been established, SQL applies predicates in the _____ clause, allowing aggregate functions to be used.

- a) ☐ Where
- b) ☒ Having
- c) ☐ Group by
- d) ☐ With

25

Answer: a

Explanation: Any property that does not occur in the group by clause must only appear in an aggregate function if it also appears in the select clause; otherwise, the query is considered incorrect.

26 Answer: b

Explanation: In SQL, after grouping data using the GROUP BY clause, the HAVING clause is used to filter the groups based on specific conditions. It allows the use of aggregate functions and selects only the groups that satisfy the given criteria.

27. What does a foreign key combined with a primary key create?
- a) Network model between the tables that connect them
 - ☒ b) Parent-Child relationship between the tables that connects them
 - c) One to many relationship between the tables that connects them
 - d) All of the mentioned

28. Which of the following is the subset of SQL commands used to manipulate Oracle Structures, including tables?

- a) Data Described Language
- b) Data Retrieval Language
- c) Data Manipulation Language
- ☒ d) Data Definition Language

27. Answer: b

Explanation: When the foreign key of one table references the primary key of another table, the foreign key creates a parent-child relationship. The table containing the primary key is considered the parent table where as the table containing the foreign key is considered the child table.

28. Answer: d

Explanation: DDLs are used to define schema and table characters and consist of commands that help in modifying. DML performs the change in the values of the relation.

27. Answer: b

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28. Answer: d

Explanation: DDLs are used to define schema and table characters and consist of commands that help in modifying. DML performs the change in the values of the relation.

29. For designing a normal RDBMS which of the following normal form is considered adequate and tell us ^{NO} Transitive Dependency comes under which NF?

- a) 4NF
- ☒ b) 3NF
- c) 2NF
- d) 5NF

30 . The oldest DB model is _____

- a) Network
- b) Physical
- ☒ c) Hierarchical
- d) Relational

29. Answer: b

Explanation: Because most 3NF tables are free of insertion, update, and deletion anomalies, an RDBMS table is sometimes regarded as “normalized” if it is in the Third Normal Form.

In 3 NF = 2 NF + ^{No} Transitive Dependency (Non prime or prime attribute access by only prime attribute)

30 . Answer: c

Explanation: Hierarchical database model was developed by IBM in the early 1960s. It had Parent-Child Relationship.

THANK YOU